Slide 1: Session opening

Main point: Turn an agent recommendation into a controlled operational decision.

The final session connects evidence, authority and execution. An answer can be well supported while the actor has no permission to act. An authorized actor can also propose a poorly supported action. We need both sets of checks and a clear owner for unresolved issues. The goal is an inspectable pilot with explicit limits, not a general statement that the model is safe.

Example to use The Harbor Metals research role can search and calculate, but cannot release a brief. A reviewer grant allows release only while it remains valid.

Assumptions and limits The examples use illustrative rules and simulated actions. Production identity, evidence checks and execution controls must be implemented and verified.

Ask the room What action should your workflow never take without a current approval?

Slide 2: The decision this session makes

Main point: Define what a release decision must include.

State the intended use, boundaries, required evidence and current authority. Then decide whether to release, restrict, escalate or continue testing. A model score alone cannot make that decision because it does not specify acceptable consequences or who owns them. Governance becomes practical when the policy produces observable permitted and blocked actions.

Example to use A pilot may be approved to draft a risk summary for review while remaining prohibited from sending it to clients.

Assumptions and limits The scope and release criteria are organizational decisions. A narrowly qualified use must not silently become a broader autonomous service.

Ask the room What is the smallest useful pilot you could authorize?

Slide 3: Roadmap: from purpose to permitted action

Main point: Explain the sequence from scope to enforcement.

Define purpose and requirements first. Map them to evidence and owners, then place a gate before protected execution. Test valid and fault cases and decide which changes require renewed review. This order prevents a technically impressive controller from enforcing the wrong policy. Every step should leave a record another person can inspect.

Example to use An action that depends on a current holdings snapshot needs both a recorded snapshot version and a check that it is still current when the order is committed.

Assumptions and limits The roadmap is a lifecycle with iteration, not a one-time checklist that guarantees every risk has been found.

Ask the room Which item currently has no named owner?

Slide 4: Five learning objectives

Main point: Describe the artifacts the session should produce.

You should be able to define the system boundary, attach requirements to evidence and owners, design a release gate, specify monitoring and decide a pilot outcome. These are practical deliverables. The questions distinguish fixed checks from sequential optimization and past approval from current authority. The exercise exposes those distinctions through permitted and blocked commits.

Example to use A governance matrix can name the rule, graph target, expert, action and record required before release.

Assumptions and limits A completed matrix is not proof the controls work. Its implementation and fault tests must support the decision.

Ask the room Which objective would close the largest gap in your existing deployment process?

Slide 5: Start with a system boundary

Main point: Explain what is inside the governed system.

List the model, prompts, data stores, retrieval, tools, controller and execution destination. Include identities, credentials and human approval channels. A guarantee about one component cannot cover a bypass outside that boundary. The boundary determines what must be versioned, tested and monitored.

Example to use If an agent can send an order directly to a broker using a separate key, a gate on the normal application path does not mediate every trade.

Assumptions and limits The boundary description must reflect real deployed paths and dependencies. Unknown integrations limit the strength of any assurance claim.

Ask the room Which credential or alternate interface could bypass the intended controller?

Slide 6: A lifecycle, not a final approval meeting

Main point: Explain why qualification can expire after change.

Governance starts when the task is defined and continues through collection, development, evaluation, deployment and monitoring. A source, prompt, model or permission can change after approval. Determine which changes invalidate which parts of the assessment and who must respond. This makes governance an operating process rather than a presentation at the end of development.

Example to use A provider updates a hosted model behind the same product name. The previous evaluation may no longer describe the actual behavior.

Assumptions and limits Version visibility and change detection must be sufficient for the proposed monitoring plan. Approval is conditional on the assessed configuration and use.

Ask the room What change would trigger a new test before continued use?

Slide 7: Policy and mechanism are different

Main point: Distinguish the intended rule from code that enforces it.

Policy states what is allowed. A mechanism implements the restriction. Asking a model not to release a brief expresses a desired behavior but does not make release impossible. A server-side permission check can block the actual release operation. Test whether the mechanism enforces the policy under normal and adversarial requests.

Example to use Policy: research may calculate but only reviewers may release. Mechanism: dispatch checks the authenticated grant before executing release_brief.

Assumptions and limits The identity and grant store must be trusted and protected. A tool hidden from the prompt can still need an execution-side check.

Ask the room Where would the system stop a prohibited request if the model ignored the prompt?

Slide 8: Two questions at every release

Main point: Keep evidential adequacy and authority independent.

First ask whether the conclusion is supported by applicable evidence and whether material gaps remain. Separately ask whether this actor may take this action under current approvals and system state. Neither side implies the other. This distinction prevents a correct answer from becoming an unauthorized action and a valid approval from becoming a factual guarantee.

Example to use A leverage calculation can be correct while release permission is absent. A reviewer can have permission while the controlling covenant document is missing.

Assumptions and limits Passing evidence checks means the declared tests passed, not that every unknown fact is true. Permission checks concern action authority.

Ask the room Which side would fail in each of those two cases?

Slide 9: Specify requirements before scoring evidence

Main point: Explain how general rules become task-specific checks.

R1 defines the decision and comparison population. R2 specifies eligible evidence and coverage. R3 checks the inference from measurements to claim. R4 checks who can authorize release. Domain owners and specialists define those requirements before the graph applies them. The graph can organize and execute specified checks; it cannot invent an organization policy from a vague request.

Example to use For a sponsorship comparison, R1 fixes the campaign audience, R2 requires comparable source populations, R3 checks the denominator and R4 requires a named approver.

Assumptions and limits Rules are illustrative and must be reviewed for the actual task. The need for evidence is defined by the decision, not a universal document count.

Ask the room Who should approve each of the four requirements?

Slide 10: Document the population behind the source

Main point: Explain evidence bias as a collection question.

Ask who was sampled, who is missing, how observations were selected and which comparisons the source supports. A large dataset can still exclude the relevant population. Missing collection metadata should remain a gap, not be converted into apparent representativeness. A tool can flag the mismatch when the target and source frames are recorded; unknown omissions still need expertise.

Example to use A fan-club survey describes its respondents, not automatically every potential customer in a sponsorship market.

Assumptions and limits Measuring selection bias generally needs a target population and information about inclusion or a justified model. Source count alone provides neither.

Ask the room What population does this source actually describe?

Slide 11: Which is better: Giants or Patriots?

Main point: Use a vague question to motivate a decision definition.

Better depends on the purpose: a fan choosing a team, a sponsor seeking an audience and an investor assessing commercial returns need different evidence. Define the metric, period and target market before ranking. The exercise is about governing the comparison, not asserting a current sporting ranking.

Example to use A sponsor interested in a particular regional audience should not substitute national fan preference for campaign return.

Assumptions and limits No team-performance facts or commercial rankings are asserted by the illustrative case. The target metric needs business approval.

Ask the room Better for whom, on what measure and over which period?

Slide 12: Map the question to its evidence and decision

Main point: Explain each graph node and edge as an explicit dependency.

O1 is the approved objective. E nodes are source records. M1 contains comparable measurements derived from those sources. C1 is the resulting claim and D1 the proposed decision. An edge records a declared dependency: which source or definition is needed for a measurement or claim. Attach requirements where they are checked so a failure has a clear downstream scope and owner.

Example to use If the two audience sources use incompatible populations, the measurement comparison is unresolved and the ranking claim should not be released as established.

Assumptions and limits The graph must be correctly constructed from source bindings and reviewed requirements. It is not automatically a causal model of consumer behavior or a complete inventory of all missing evidence.

Ask the room Which node would you inspect if the denominator differs between the two sources?

Slide 13: The governance matrix: checks, experts and decisions

Main point: Explain how the graph becomes a leadership decision record.

Read each row as a mapping from a requirement to a concrete check, a responsible expert and a required record. Business owners define objectives, data specialists inspect collection and comparability, relevant experts review inference and authorized approvers own release. The graph can populate status and affected objects, while the policy determines which action follows. This avoids an unexplained red score without an operational next step.

Example to use R2 can flag that only one team has an eligible audience source, route it to the data steward and hold the comparison until the gap is repaired or the claim narrowed.

Assumptions and limits Actions and routing rules must be approved beforehand. Automated detection does not confer expertise or authority.

Ask the room Which record would demonstrate that the responsible expert actually resolved the issue?

Slide 14: So what decision does the graph support?

Main point: Explain clarify, escalate, withhold and release as different outcomes.

If the objective is unclear, clarify it. If a technical or domain concern needs judgment, escalate it to the relevant owner. If a material support gap remains, withhold the affected claim. Release a bounded conclusion only after requirements and approval are met. These actions can be scoped to one claim rather than reject an entire document.

Example to use An audience count can remain reportable while a comparative return-on-sponsorship ranking is withheld because the inference is unsupported.

Assumptions and limits The impact scope depends on correct graph dependencies and approved policy. Release retains stated limitations rather than implying complete certainty.

Ask the room What can still be published when only the comparative claim is unresolved?

Slide 15: What the graph contributes

Main point: Describe concrete value beyond a list of warnings.

Typed records identify sources, facts, measurements, claims and actions. Dependencies show which objects use which inputs. Rule attachments identify where requirements are tested. Repair scope identifies which descendants need rechecking after a change. These features make an issue traceable to a decision and an owner. They do not eliminate the need for domain judgment.

Example to use A corrected source updates the relevant measurement and comparison claim, while unrelated descriptive facts remain unchanged.

Assumptions and limits Completeness of the dependency record is an assumption to verify. A graph drawn after the fact can be misleading if it does not reflect actual executed bindings.

Ask the room What does the edge mean, and how would you verify it?

Slide 16: Question: many sources, one population

Main point: Check whether source volume is being confused with coverage.

One hundred fan-club surveys can contain substantial information about that club while still failing to represent the intended wider audience. Ask participants which response addresses the missing population rather than the impressive document count. A model opinion that the evidence looks balanced does not establish sampling coverage.

Example to use Duplicating observations from the same selected group does not add the missing groups to the dataset.

Assumptions and limits The target population and source inclusion process must be known enough to assess the mismatch. The example is hypothetical.

Ask the room Which option checks the right denominator?

Slide 17: Answer: C — check coverage

Main point: Explain what an expert needs to resolve representativeness.

C is correct. Record the difference between the source population and the target decision population, identify the affected claim and route it to a specialist. A remedy can be a suitable additional source, a justified adjustment or a narrower claim. Do not apply a generic correction without knowing how collection and inclusion work.

Example to use Limit the statement to surveyed fan-club members if that is all the source supports, rather than present it as a conclusion about the whole sponsorship market.

Assumptions and limits Unknown selection cannot be fully corrected from source counts alone. Any adjustment needs assumptions and validation.

Ask the room When would narrowing the claim be preferable to collecting more data?

Slide 18: From a check to an operational state

Main point: Explain why resolution, approval and release are separate.

A proposal can be held while evidence is repaired, then approved by an accountable actor and only afterward released. If evidence or authority changes, return it to review. A state record prevents an earlier approval from being treated as permanent permission despite changed conditions. Keep each transition and its reason in the audit trail.

Example to use A brief is approved using version one of a covenant, but a controlling update arrives before release. The pending release should be reassessed.

Assumptions and limits State transitions and version binding must be implemented consistently. A state label alone does not enforce the underlying checks.

Ask the room Which change should move an approved proposal back to hold?

Slide 19: Place enforcement at the action boundary

Main point: Explain complete mediation in practical terms.

The agent proposes an action. The controller checks it against policy and current state. Only the permitted request reaches the executing tool. Every protected route must pass that controller; otherwise the agent can bypass the gate through another endpoint or credential. The model need not be malicious for this to matter: a mistaken or repeated request is enough.

Example to use A research assistant requests release_brief. The tool server checks its current grant before any document is sent.

Assumptions and limits The controller, identity, policy and execution path form a trusted boundary. A proof covering one path does not cover an unmodeled bypass.

Ask the room What alternate call path would invalidate the guarantee?

Slide 20: A commit-time predicate

Main point: Explain the conjunction as four required yes-or-no checks.

Allow returns true only if all four components are true. E means the declared evidence checks pass for this proposal. P means current authority covers the action. V means the source and state versions still match. U means the operation has not already been committed. Recheck immediately before the side effect, not only when the proposal was drafted. This is a logical eligibility rule, not a probability model.

Example to use A proposal has good evidence and permission but has already been committed once. U is false, so a repeated request is blocked.

Assumptions and limits The predicates must be correct, identity and state trusted, versions bound to the proposal and check-plus-execution protected against intervening changes. Failing closed on unknown status is an explicit policy choice.

Ask the room Which predicate changes when permission is revoked?

Slide 21: What could actually be proved?

Main point: State a narrow invariant together with its assumptions.

A formal claim might say no committed action exceeds the valid grant represented in the system. Such a claim depends on complete mediation, correct checks, protected policy and atomic handling of state. It does not prove source truth, future profitability or absence of every unknown threat. Finite replay tests provide empirical evidence about selected cases; they are not a proof over every execution.

Example to use If all tool calls use the controller but the agent also holds an independent broker key, the modeled invariant no longer covers every possible trade.

Assumptions and limits The system model must include actual credentials and execution paths. Specification errors, implementation bugs and unmodeled behavior are outside a proof that assumes them away.

Ask the room What exact sentence would you be willing to guarantee, and under which assumptions?

Slide 22: Question: approved at 08:55

Main point: Check the time-of-check versus time-of-use issue.

The proposal was approved before the grant was revoked. Execution happens afterward. Ask whether an earlier approval is enough to authorize the later action. It is not under a policy requiring current authority. The controller must consult the grant state at execution and bind it to the specific proposal.

Example to use The same release request is valid at 08:55 and invalid at 09:00 after revocation. No change to the model answer is needed for permission to change.

Assumptions and limits The scenario assumes revocation is effective immediately and visible to the controller. Propagation delays must be included in a production policy.

Ask the room Which component should make this decision rather than ask the language model?

Slide 23: Answer: B — recheck at execution

Main point: Explain why saved approval is historical evidence, not permanent authority.

B is correct. The earlier approval records a past decision. The current grant determines whether the protected action may happen now. Block the operation after revocation and record why, even if the text of the brief and recommendation is unchanged. This is an execution control, not a preference-learning task.

Example to use The Harbor notebook uses the same brief for the permitted reviewer call and the later revoked call. The difference is the current grant.

Assumptions and limits Identity and revocation state must be trustworthy and checked atomically enough to prevent a race between approval and execution.

Ask the room What record would let an auditor distinguish a revoked grant from a content-quality rejection?

Slide 24: When do MDPs become relevant?

Main point: Explain when fixed rules are enough and when actions affect future choices.

A known prerequisite can be checked with a predicate. A state machine tracks approvals, retries and commits. An MDP becomes useful when choosing actions changes later states and consequences. A POMDP is relevant when important state is hidden and observations update a belief about it. Do not use sequential optimization merely to make a rule sound sophisticated. It needs a decision problem with a useful transition and cost model.

Example to use Blocking an expired grant is a fixed check. Deciding whether another investigation is worth its delay and expense can require comparing future information and consequences.

Assumptions and limits An MDP assumes an adequate Markov state and specified transition and reward model. A POMDP also needs an observation model and belief update. Real systems may violate or only approximate these assumptions.

Ask the room Does the action change what you can know or do next?

Slide 25: A decision policy still needs hard limits

Main point: Explain the policy-value equation without confusing optimization with authority.

A policy selects an allowed action in each state. Its value is the expected sum of rewards over the stated horizon, discounted by gamma. The controller defines which actions are allowed; optimization chooses among them. A high expected reward cannot authorize an otherwise prohibited action. A language-model agent can participate in this sequential framework without having been trained through reinforcement learning.

Example to use Investigation, escalation and no action can be allowed choices. Sending a client instruction can remain outside the grant even if the estimated reward appears high.

Assumptions and limits The displayed finite-horizon expectation assumes defined states, transitions, rewards and a policy with integrable returns; gamma is in zero to one. A stationary Markov formulation needs sufficient state, or time included in state. Hidden state requires a belief model. No i.i.d. trajectory-step assumption is made.

Ask the room Which part defines authority and which part optimizes consequences?

Slide 26: Question: which needs sequential reasoning?

Main point: Distinguish eligibility checks from value-of-information choices.

An expired grant and a required signature can usually be tested directly once the relevant facts are available. Choosing whether to investigate more is different: that action consumes resources and can change the information available for a later decision. Ask participants to explain why C is structurally different, not merely more complicated in wording.

Example to use A second source might resolve an uncertainty, but obtaining it could delay a time-sensitive decision. Its usefulness depends on what we expect to learn and what delay costs.

Assumptions and limits A sequential model helps only if its states, outcomes and costs can be specified meaningfully. A simple approved rule may still be adequate.

Ask the room What future consequence does investigation change?

Slide 27: Answer: C — value of more information

Main point: Explain the cost-benefit comparison behind another inquiry.

C is the sequential choice. Compare the expected improvement in the eventual decision after investigation with the cost and delay of obtaining information. A and B can be enforced as prerequisites. The distinction does not mean an agent may override those prerequisites while investigating. Hard limits remain outside the optimizer choices.

Example to use A low-cost extra document may be worth retrieving when it can reverse a material release decision; it may not be worth delaying an irrelevant administrative answer.

Assumptions and limits Expected value calculations depend on a credible uncertainty and consequence model. They do not make the newly retrieved source automatically reliable.

Ask the room When would you stop investigating and explicitly withhold?

Slide 28: Test the controller as well as the model

Main point: Specify independent expected outcomes for valid and fault cases.

Test a valid proposal that should be permitted and faults that should be blocked: missing evidence, changed versions, revoked grants and duplicates. Measure both false releases and unnecessary holds. Expected outcomes should come from the approved policy and review process, not from the gate being tested. Keep held-out combinations so the controller is not only demonstrating its own handpicked examples.

Example to use A test record contains principal=research, action=release, evidence complete=true, grant permits release=false and expected result=deny. A reviewer-grant variant can be the valid control.

Assumptions and limits These are normative policy labels, not future-profit labels. Fault prevalence and correlated case families affect empirical rate estimates. Passing finite tests is not a universal proof.

Ask the room What valid case prevents an always-block gate from looking perfect?

Slide 29: Monitor what invalidates qualification

Main point: Explain monitoring as preservation of the assessed conditions.

Watch source completeness and coverage, tool failures, missing traces, behavior changes and versions of the model, prompt, corpus and grants. Outcome labels may arrive later, so early pipeline checks and delayed performance checks serve different purposes. Approval should identify which changes require review and who responds.

Example to use A source feed stops providing one region while the system keeps producing summaries. The missing coverage can invalidate the supported scope before outcome metrics visibly deteriorate.

Assumptions and limits Alerts need defined thresholds, owners and response procedures. A distribution-shift statistic alone does not establish a business failure or its cause.

Ask the room Which alert would trigger a hold rather than merely a dashboard notification?

Slide 30: A trigger starts diagnosis, not automatic retraining

Main point: Explain why different failures need different repairs.

A missing source calls for ingestion or retrieval repair. A broken tool schema calls for integration work. A behavior failure with valid context can motivate prompt or training changes. A changed business scope may require new approval rather than a model update. Diagnose the component, repair it and rerun the affected evaluation. Automatic retraining can preserve or amplify an evidence problem.

Example to use If a calculator argument changed units from millions to dollars, new demonstrations are not the first remedy; fix and validate the interface.

Assumptions and limits The diagnosis must be supported by controlled tests. Changes can interact, so regression checks should cover dependent and unaffected branches.

Ask the room Which observed trigger in your system should never directly launch retraining?

Slide 31: Class Exercise: authorization must survive until execution

Main point: Guide the web exercise through a valid control and injected faults.

Prepare an approved proposal and commit it under unchanged conditions. Repeat the commit to see the duplicate check. Reset, prepare a fresh proposal, change the state and try again. Inspect the failed predicate in the audit trail. Repeat with revocation or changed pending orders. The contrast isolates a changing execution condition while the original proposal stays fixed.

Example to use An order computed without a newly pending order can be blocked because the decision state no longer matches the current state, even if its original model reasoning was sensible.

Assumptions and limits The exercise simulates execution. Its permission and version model must be distinguished from a production broker guarantee. The valid control is essential: a system that blocks everything does not satisfy the intended policy.

Ask the room Ask participants to point to the exact condition that failed, not say the AI was unsafe.

Slide 32: Design your own governed pilot

Main point: Turn the concepts into one limited operating plan.

Choose one workflow and name its user, inputs, output, prohibited actions and accountable owner. Define two blocking requirements and an escalation path. Write one valid case and two fault cases with expected outcomes. Name records and change triggers. Present a release decision under one fault and explain the supporting evidence. This makes the governance design concrete enough to review.

Example to use A research drafting pilot can permit source lookup and calculation while prohibiting publication until a reviewer approves the bound source version.

Assumptions and limits The pilot is deliberately scoped and must not be presented as evidence of general enterprise reliability. Policy labels need owner approval.

Ask the room What is the smallest useful action the pilot can safely support?

Slide 33: What the pilot must produce

Main point: Explain why each artifact supports a different objective.

The scope record defines what is being qualified. The requirements matrix links checks and owners. The gate replay shows observable behavior on valid and fault cases. The monitoring plan says when qualification needs review. The final decision records whether to release, restrict or continue testing and why. These pieces let leadership inspect both value and limits.

Example to use A decision can approve analyst drafting but withhold autonomous external release until identity and evidence-binding controls are validated.

Assumptions and limits Documentation is necessary but not sufficient. Records must correspond to functioning mechanisms and independently reviewed tests.

Ask the room Which artifact would be most difficult to produce today?

Slide 34: Reading map: governance and enforcement

Main point: Use established security concepts to sharpen the deployment claim.

Policies, mechanisms, complete mediation and runtime control predate language models. They remain useful because an agent action still reaches an ordinary software interface. Use these references to specify what every protected call must satisfy and which trusted components enforce it. General AI risk frameworks add lifecycle and accountability structure. Neither replaces concrete implementation evidence.

Example to use Complete mediation means every release path checks authority, not just the normal path in the demo.

Assumptions and limits The references have their own formal models. A guarantee cannot be transferred by analogy without showing that the deployment satisfies the assumptions.

Ask the room Which unmodeled bypass would a security review ask about first?

Slide 35: Reading map: decisions, data and monitoring

Main point: Explain which sources support the remaining operating questions.

Sequential-decision material helps when investigation changes future options. Dataset documentation helps assess source populations and suitable uses. Monitoring material helps preserve the conditions under which the system was assessed. Select the source according to the unresolved decision rather than treat the reading list as a compliance checklist.

Example to use If a population is missing, investigate collection and scope before optimizing a release threshold. If another inquiry has uncertain value, specify its possible observations and costs.

Assumptions and limits Each modeling tool simplifies reality. Record the simplifications and test whether they are adequate for the bounded pilot.

Ask the room Which unresolved question belongs to a domain expert rather than a model score?